[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, UNIVERSAL, VERTICAL-SPINDLE, NONTRAVERSING ROTARY TABLE, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC), nontraversing rotary table, vertical-spindle, universal grinding machines intended for use in shops for finish grinding of internal and external surfaces, tapers, and shoulders on workpieces with rapid metal removal rates to produce precision flat surfaces on a variety of materials.

2. CLASSIFICATION. The grinding machines shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Adjustable crossrail
Type II - Fixed crossrail

Style A - Single wheelhead
Style B - Dual wheelheads

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Grinding machines sizes.

Size	Worktable diameter, inches	Swing, inches	Bore length, inches
1	16	18	12
2	20	22	12
3	24	20	20
4	24	24	12
5	24	36	18
6	30	35	18
7	36	44	18
8	42	50	18
9	48	60	24
10	60	68	24

Note: The range of machine sizes shown in table I imply that the machine offered shall be equal to or larger than the size stated but smaller than the next available size. When the largest machine size is ordered, the machine offered shall not exceed the stated requirements by more than 10 percent.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the grinding machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the grinding machine.

3.2.2 Worktable. Unless otherwise specified (see 7.2(c)), the diameter, rotational speed range, worktable motor horsepower, and weight-holding capacity of the worktable shall be that considered by the manufacturer as standard. The number and size of T-slots shall be as specified (see 7.2(d)). The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1.

3.2.3 Worktable spindle. The base shall support the worktable spindle. Unless otherwise specified (see 7.2(e)), the worktable spindle nose shall be the manufacturer's standard design for the attachment of the rotating worktable or faceplates. The worktable spindle shall have infinitely variable rotational speed.

3.2.4 Column(s). One or more vertical columns shall be provided to support the crossrail. Type I machine columns shall allow vertical (Z-axis) movement of the crossrail.
3.2.5 Crossrail. The crossrail shall support and allow horizontal (X-axis) movement of the cross-slide.

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3.2.6 Cross-slide. Unless otherwise specified (see 7.2(f)), the cross-slide shall support and move the grinding wheelhead assembly along the crossrail horizontal X-axis to allow grinding over the full worktable diameter.

3.2.7 Grinding wheelhead assembly. Unless otherwise specified (see 7.2(g)), the grinding wheelhead assembly shall be the design and have the drive motor horsepower considered by the manufacturer as standard for the machine. Unless otherwise specified (see 7.2(h)), both grinding wheelheads of a style B machine shall have identical characteristics.

3.2.8 Grinding wheelhead swivel. Unless otherwise specified (see 7.2(i)), the grinding wheelhead shall swivel ± 45 degrees left and right from the vertical center. The grinding wheelhead swivel shall be by manual adjustment or the CNC system as specified (see 7.2(j)). When manual adjustment is provided, the wheelhead shall have a scale divided in increments of 1 degree or less to indicate swivel position and shall have a lock to hold swivel adjustment.

3.2.9 Machineways. Unless otherwise specified (see 7.2(k)), machineways for the column, cross-slide, and grinding wheelhead assembly shall be of the design considered by the manufacturer as standard for the grinding machine. All machineways shall have protective covers and/or wipers to keep chips, coolant, and other debris from bearing surfaces.

3.2.10 Wheel dresser. Unless otherwise specified (see 7.2(l)), an adjustable wheel dresser, complete with diamond point and capable of dressing the face and periphery of the grinding wheel, shall be provided.

3.2.11 Grinding wheel changer and storage magazine. When specified (see 7.2(m)), an automatic grinding wheel changer and storage magazine shall be provided. When provided, the wheel changer shall have manual or CNC system control as specified (see 7.2(n)). The grinding wheel changer shall select the next grinding wheel from the storage magazine using the bidirectional random shortest path method, exchange grinding wheels, return the previous grinding wheel to the storage magazine, and track the grinding wheel for reuse.

3.2.12 Grinding wheels. Grinding wheels shall be supplied with the grinding machine as specified (see 7.2(o)). Unless otherwise specified (see 7.2(p)), grinding wheels shall conform to ANSI B74.2, be marked in accordance with ANSI B74.13, and meet safety requirements of ANSI B7.1.

3.2.13 CNC system. The CNC system shall be based on either a personal computer (PC) open-architecture or other software-programmable microprocessor design. The CNC system shall be industrial grade in accordance with accepted industry practices and shall meet Government requirements as specified (see 7.2(q)).

3.2.13.1 Hardware features. Unless otherwise specified (see 7.2(r)), the CNC system shall include, as a minimum, the following hardware:

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a. One or more 32-bit microprocessors with math coprocessor.
b. 250-MB hard disk drive.
c. An ANSI/Telecommunications Industry Association/Electronic Industries Association (TIA/EIA)-232 communications port.
d. 9-inch monochrome monitor.
e. Manual data input (MDI) device: PC keyboard or alphanumeric keyboard with dedicated keys for specific machine functions.

3.2.13.2 Control panel. Unless otherwise specified (see 7.2(s)), the CNC system shall include, as a minimum, a control panel with the following manual controls and switches:

a. System keyed lock.
b. Emergency stop button.
c. Power on/off switches.
d. CNC system start/stop switches (cycle start/stop switches).
e. Spindle start/stop switches; spindle motor power and spindle speed meters.
f. Spindle speed override rotary switch.
g. Feed hold.
h. Feed rate override rotary switches.
i. Jog and jog direction.

3.2.13.3 Operating modes. Unless otherwise specified (see 7.2(t)), the CNC system shall include, as a minimum, the following operating modes:

a. Manual operation mode.
b. Manual data input mode.
c. Setup mode.
d. Learn mode.
e. Transmit and receive mode.
f. Automatic mode.
g. Grinding machine diagnostic/maintenance/monitoring modes.

3.2.13.4 Axis and tool management. Unless otherwise specified (see 7.2(u)), the CNC system shall include, as a minimum, the following axis and tool management features:

a. Grinding wheel diameter compensation.
b. Grinding wheel breakage detection.
c. Scale feedback.
d. Grinding wheel offsets.
e. Number of grinding wheel offsets.
f. Grinding wheel dresser cycle.

3.2.13.5 Software features. Unless otherwise specified (see 7.2(v)), the CNC system shall include, as a minimum, the following software programming and control features:

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a. Linear (X-axis and Z-axis) control of the wheelhead and simultaneous control of the worktable rotational speed and direction (A-axis).
b. Linear and circular interpolation.
c. Inch/metric switchable programming.
d. Decimal point programming.
e. Workpiece graphic display rotation.
f. Tool path graphic display.
g. Reference zero.
h. Standard G codes, M codes, T codes, and canned cycles.
i. Dimensional absolute/incremental programming.
j. Cartesian/polar coordinate programming.

3.2.14 Lubrication system. Unless otherwise specified (see 7.2(w)), the grinding machine shall be provided with an automatic lubricating system to direct oil flow from the reservoir to all machine components requiring continuous lubrication. A means shall be provided for determining oil level in the reservoir. The automatic lubricating system shall include a cleanable or replaceable filter. Upon loss of pressure in the automatic lubrication system, the grinding machine shall automatically stop. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.15 Coolant system. Unless otherwise specified (see 7.2(x)), the grinding machine shall be provided with a flood coolant system to control the rate of coolant flow from the reservoir and direct coolant flow to the grinding area. As a minimum, the coolant system shall be complete and include a reservoir, pump, strainer, fluid level indicator, and piping. When specified (see 7.2(y)), an optional filtration system shall be provided.

3.2.16 Hydraulic system. If a hydraulic system is provided, it shall conform to the requirements of National Fluid Power Association (NFPA) T2.24.1. The hydraulic system shall be complete and include a reservoir, pump, valves, piping, cylinders, fluid level indicator, and controls. Means shall be provided to prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid.

3.2.17 Electrical system. Unless otherwise specified (see 7.2(z)), the electrical system shall conform to the National Fire Prevention Association (ANSI/NFPA) 79. The grinding machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(aa)). Unless otherwise specified (see 7.2(bb)), the CNC system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the grinding machine main power supply.

3.2.18 Electric motors. The grinding machine drive motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1. Unless otherwise specified (see 7.2(cc)), the motors shall have open drip-proof enclosures.

3.2.19 Optional accessories. Optional accessories shall be provided as specified (see 7.2(dd)).

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3.3 Performance. Unless otherwise specified (see 7.2(ee)), the grinding machine shall meet the alignment and performance requirements shown in tables II and III, respectively, when tested in accordance with section 5.3 through 5.4.

TABLE II. Alignment requirements.

Parameter	Performance
Rotary worktable top surface flat in all directions	
- up to 60-inch worktable diameter	0.0003 inch/12 inches TIR[1]
- 61-inch and larger worktable diameter	0.0005 inch/12 inches TIR[1]
Rotary worktable top and periphery surfaces, axial runout	0.0002 inch TIR[1]
Cross-slide parallel with rotary worktable top	0.0005 inch/12 inches TIR[1]
Positioning vertical travel of grinding wheelhead, straight	0.001 inch/12 inches TIR[1]
Grinding wheelhead vertical travel during grinding, straight	0.0003 inch/12 inches TIR[1]
Grinding wheelhead spindle travel square with rotary worktable surface	0.0005 inch/12 inches TIR[1]
Grinding wheelhead spindle square with surface of rotary worktable	0.0003 inch/12 inches TIR[1]
Taper of wheel spindle true running, axial runout	0.0003 inch TIR[1]
Wheel and worktable spindles, axial runout	0.0003 inch TIR[1]

[1] Total indicated reading.

TABLE III. Performance requirements.

Parameter	Performance
Diameter	0.0003 inch TIR[1]
Straight	0.0002 inch/foot TIR[1]
Surface finish	8 microinches R_t[2]

[1] Total indicated reading.

[2] R_t is total roughness as defined in American Society of Mechanical Engineers (ASME) B46.1, with an evaluation length of 0.1 inch.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(ff)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health

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Administration (OSHA) 29 CFR PART 1910. If a conflict arises between ANSI B11.9 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(gg)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(hh)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(ii)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(jj)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

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5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(kk)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(ll)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(mm)). Unless otherwise specified (see 7.2(nn)), the grinding machine shall be tested in an indoor facility with the temperature between 40 and 110 degrees Fahrenheit and the relative humidity between 20 and 95 percent, noncondensing.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operational test procedures. The grinding machine shall be operated at no load for not less than 4 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. The grinding machine alignment tests shall be conducted before and after the performance tests and shall meet the requirements shown in table II. Changes in alignment shall be cause for rejection.

5.3.4 Performance tests. The manufacturer shall demonstrate the grinding machine capability to finish grind the internal (A), external (C), taper (B and E), and shoulder (D) surfaces of the test workpiece.

5.3.4.1 Grinding test workpiece. The test workpiece shown in figure 1 shall be of AISI 1045 steel with a minimum 6-inch diameter bore. The bore length (F) shall be equal to the rated bore length shown in table I for the size of the grinding machine provided.

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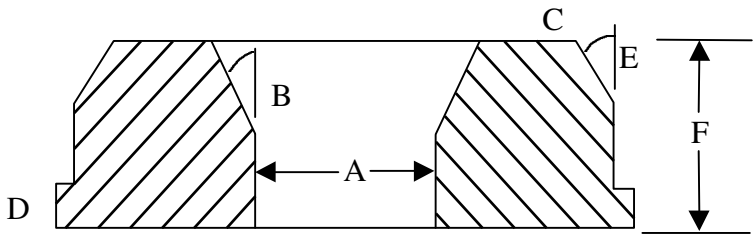

Surface A: Internal straight bore diameter.
Surface B: Internal 30 degree taper.
Surface C: Face.
Surface D: External shoulder.
Surface E: External 30 degree taper.
Surface F: Bore length.

FIGURE 1. Sectional view of test workpiece.

5.3.4.2 Grinding test. The test workpiece shall be ground in one setup and meet the performance requirements shown in table III. The test workpiece shall have 0.005 inch ground from all surfaces. The following grinding steps shall be completed in any order:

a. Grind a straight internal bore of the diameter determined by the manufacturer (not less than 6 inches).
b. Grind a 30-degree internal taper.
c. Grind a 30-degree external taper.
d. Face the end of the workpiece.
e. Grind an external shoulder of a size determined by the manufacturer.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(oo)).

5.4 Acceptance. Unless otherwise specified (see 7.2(pp)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables II and III, respectively. Failure of the grinding machine to meet the requirements shown in tables II and III shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(qq)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

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7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B5.1	T-Slots, Their Bolts, Nuts, and Tongues
ANSI B7.1	Safety Requirements for the Use, Care and Protection of Abrasive Wheels
ANSI B11.9	Safety Requirements for the Construction, Care and Use of Grinding Machines
ANSI B74.2	Specification for Shapes and Sizes of Grinding Wheels and for Shapes, Sizes and Identification of Mounted Wheels
ANSI B74.13	Markings for Identifying Grinding Wheels and Other Bonded Abrasives

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

Electronic Industries Association (EIA)

ANSI/TIA/EIA 232	Interface Between Data Terminal Equipment and Data Circuit-Termination Equipment Employing Serial Binary Data Interchange

Applications for copies should be sent to Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

National Electrical Manufacturers Association (NEMA)

ANSI/NEMA MG 1	Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

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National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P. O. Box 9101, Quincy, MA 02269-9101.

National Fluid Power Association (NFPA)

NFPA T2.24.1 Hydraulic Fluid Power - System Standard for Stationary Industrial Machinery First Edition

Applications for copies should be sent to the National Fluid Power Association, 3333 North Mayfair Road, Suite 311, Milwaukee, WI 53222-3219.

U. S. Department of Labor, Occupational Safety and Health Administration (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR 1910)

Applications for copies should be sent to the U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, style, and size of grinding machine required (see 2).
c. Worktable diameter, rotational speed range, motor horsepower, and capacity, if different (see 3.2.2).
d. Number and size of T-slots, as specified (see 3.2.2).
e. Worktable spindle nose, if different (see 3.2.3).
f. Cross-slide, if different (see 3.2.6).
g. Grinding wheelhead assembly, if different (see 3.2.7).
h. Style B grinding wheelhead characteristics, if different (see 3.2.7).
i. Grinding wheelhead swivel, if different (see 3.2.8).
j. Grinding wheelhead swivel control, as specified (see 3.2.8).
k. Machineways, if different (see 3.2.9).
l. Wheel dresser, if different (see 3.2.10).
m. Grinding wheel changer and storage magazine, if required (see 3.2.11).
n. Wheel changer control, as specified (see 3.2.11).
o. Grinding wheels, as specified (see 3.2.12).
p. Grinding wheel standards, if different (see 3.2.12).
q. CNC system requirements, as specified (see 3.2.13).
r. CNC hardware features, if different (see 3.2.13.1).
s. Control panel, if different (see 3.2.13.2).

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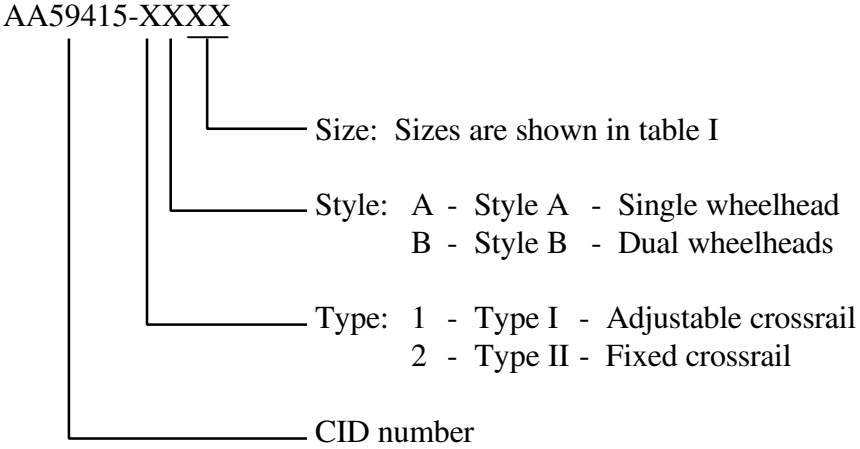
t. CNC operating modes, if different (see 3.2.13.3).
u. CNC axis and tool management features, if different (see 3.2.13.4).
v. CNC software features, if different (see 3.2.13.5).
w. Lubrication system, if different (see 3.2.14).
x. Coolant system, if different (see 3.2.15).
y. Coolant system filter, if required (see 3.2.15).
z. Electrical system industrial standard, if different (see 3.2.17).
aa. Electrical system voltage, as specified (see 3.2.17).
bb. Control system voltage, if different (see 3.2.17).
cc. Motor enclosures, if different (see 3.2.18).
dd. Optional accessories, as specified (see 3.2.19).
ee. Performance, if different (see 3.3).
ff. Machine dimensions and weight, as specified (see 3.4).
gg. System of units required (see 3.6).
hh. Dual system of units, if required (see 3.6).
ii. Nameplate, if different (see 3.7).
jj. Lubrication plate or chart, if required (see 3.8).
kk. Testing, if different (see 5.3).
ll. Test responsibility, if different (see 5.3.1).
mm. Certification of calibration, as specified (5.3.1).
nn. Test environmental conditions, if different (see 5.3.1).
oo. Optional tests, as specified (see 5.3.5).
pp. Acceptance, if different (see 5.4).
qq. Packaging requirements, if different (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor



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MILITARY INTERESTS:

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Air Force - 84
Army - AR
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PREPARING ACTIVITY:
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